Huawei Access Node HEB-ACC-42 Routing Protocol Flap Resolution SOP

Node: HEB-ACC-42 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_4229

Description

The Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure domain is a critical issue that can impact the stability and reliability of the network. In the context of the Huawei Access Node HEB-ACC-42, this failure domain is primarily caused by the dynamic nature of routing protocols, which can lead to frequent route flapping, OSPF adjacency loss, or ISIS LSP flooding. These events can be triggered by various factors, including misconfigured routing policies, faulty hardware, or software bugs. The Huawei Access Node, which is responsible for providing access to the network, plays a crucial role in propagating routing information and maintaining network connectivity. The BGP route churn, OSPF adjacency loss, and ISIS LSP flooding can occur due to the complex interactions between the routing protocols and the underlying network topology, making it essential to employ a systematic approach to diagnose and resolve this issue.

Pre-requisite Checks

- check: Verify that the Hebbal Edge site is accessible and the Huawei Access Node HEB-ACC-42 is powered on and functioning correctly.

- check: Check the status of the redundant links and ensure that they are operational and not contributing to the failure domain.

- check: Obtain the necessary permissions and access to the Huawei Access Node HEB-ACC-42 for troubleshooting and resolution.

Potential Root Causes

Cause	Probability
Misconfigured routing policies causing BGP route churn.	High
Faulty hardware or software bugs leading to OSPF adjacency loss.	Medium
Complex network topology causing ISIS LSP flooding.	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table, Expected Output: Output showing BGP routing table and any route flapping., Description: This command displays the BGP routing table and helps identify any route flapping that may be contributing to the failure domain.

Command: display ospf neighbor, Expected Output: Output showing OSPF neighbor status and any adjacency loss., Description: This command displays the OSPF neighbor status and helps identify any adjacency loss that may be contributing to the failure domain.
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```
Command: display isis lsp-list, Expected Output: Output showing ISIS LSP list and any flooding., Description: This command displays the ISIS LSP list and helps identify any flooding that may be contributing to the failure domain.
```

Resolution Steps

Step 1: Step: Reset BGP peer., Command: reset bgp peer force, Description: This command resets the BGP peer and helps resolve route flapping caused by misconfigured routing policies.

Step 2: Step: Check and replace faulty hardware., Command: check hardware, Description: This command checks the hardware and helps identify any faulty components that may be contributing to OSPF adjacency loss.

Step 3: Step: Optimize network topology., Command: optimize network topology, Description: This command optimizes the network topology and helps resolve ISIS LSP flooding caused by complex network interactions.

Post-Check Verification

```
Command: display bgp routing-table, Expected Output: Output showing stable BGP routing table., Description: This command verifies that the BGP routing table is stable and free from route flapping.

Command: display ospf neighbor, Expected Output: Output showing stable OSPF neighbor status., Description: This command verifies that the OSPF neighbor status is stable and free from adjacency loss.

Command: display isis lsp-list, Expected Output: Output showing stable ISIS LSP list., Description: This command verifies that the ISIS LSP list is stable and free from flooding.
```

Roll-back Procedures

- Step: Restore original configuration., Command: rollback configuration, Description: This command restores the original configuration and reverses any changes made during the resolution process.

- Step: Reset BGP peer., Command: reset bgp peer force, Description: This command resets the BGP peer and restores the original state.

- Step: Verify network stability., Command: display bgp routing-table, Description: This command verifies that the network is stable and free from route flapping.